When Fine-Tuning Hurts: Failure Modes of Visuomotor Imitation Learning on a Low-Cost Robot

Heyang Huang
Department of Computer Science
University of Colorado Boulder
United States
Heyang.Huang@colorado.edu

Chen Liu
Department of Computer Science
University of Colorado Boulder
United States
Chen.Liu@colorado.edu

Nick Jackoski
Department of Computer Science
University of Colorado Boulder
United States
Nick.Jackoski@colorado.edu

Nikhil Sawane
Department of Robotics
University of Colorado Boulder
United States
Nikhil.Sawane@colorado.edu

Xavier O'Keefe
Ann H.J. Smead Department of Aerospace Engineering Sciences
University of Colorado Boulder
United States
Xavier.Okeefe@colorado.edu

Abstract: Imitation learning can enable robotic manipulation by learning policies directly from demonstrations, but real-hardware deployment remains brittle under limited data, partial observability, and distribution shift [1]. We present a feasibility study and failure analysis of visuomotor imitation learning for a tabletop pick-and-place task on XLeRobot, a low-cost open-source platform. Using the LeRobot pipeline [2], we evaluate a pretrained vision-language-action policy (SmolVLA) [3] and fine-tuned variants trained on a small, success-only teleoperation dataset. While the pretrained policy exhibits reasonable reaching behavior, fine-tuning frequently degrades performance into degenerate motions such as repeated downward reaching. We attribute these failures to occlusion-driven ambiguity from monocular RGB sensing, noisy and viewpoint-misaligned demonstrations, and weak grounding between language prompts and sensorimotor feedback. Rather than demonstrating task mastery, this study surfaces practical failure modes and provides actionable guidance for improving real-robot fine-tuning through better observability, data quality control, and recovery-oriented supervision.

Keywords: imitation learning, failure analysis, fine-tuning, visuomotor policy, real-robot

1 Introduction

Robotic manipulation in real-world environments requires policies that operate reliably under uncertainty and partial observability. Even for a simple pick-and-place task, distributional shift between training demonstrations and deployment conditions (e.g., viewpoint changes, occlusions, and lighting variation) can cause imitation learning policies to fail catastrophically. Classic behavior cloning is particularly sensitive to such shift due to compounding error [1], motivating careful analysis of what breaks in real systems.

Recent progress in vision-language-action (VLA) policies has enabled end-to-end visuomotor control conditioned on natural language instructions [4]. These models can produce surprisingly competent behavior out-of-the-box, but it remains unclear when and how small-scale fine-tuning on real-robot demonstrations improves performance versus when it can *harm* it. This question matters for low-cost platforms and course/lab settings where datasets are small and sensors are limited.

In this work, we study a practical setting: fine-tuning a pretrained VLA policy (SmolVLA) [3] for a pick-and-place task on XLeRobot using a small success-only dataset collected via teleoperation. We report qualitative performance of the base policy and fine-tuned checkpoints, and we identify recurrent failure modes. Our main contribution is not a new algorithm, but a concrete feasibility + diagnostic study that clarifies the conditions under which fine-tuning can regress into degenerate behavior, and what instrumentation/data changes are most likely to fix it.

Contributions

We make three contributions: (i) an end-to-end real-robot training and evaluation pipeline on a low-cost platform using LeRobot [2]; (ii) a qualitative comparison showing that fine-tuning can degrade behavior relative to a pretrained baseline in low-data, occlusion-heavy settings; (iii) a structured failure analysis connecting observed behaviors to observability limits, demonstration quality, and language-action grounding.

2 Related Work

Imitation learning and behavior cloning learn policies from expert demonstrations and have enabled end-to-end visuomotor control in various domains [5, 6]. However, standard supervised imitation learning is known to suffer from distributional shift and compounding error when the policy visits states not seen in the demonstration distribution [1].

In recent years, Vision-Language-Action (VLA) models have shown promise in robotics by extending visuomotor policies with language-conditioned representations. By incorporating natural language into the policy input, these models enable more generalizable and reusable behaviors across a range of tasks. These models leverage large scale vison-language pre-training to interpret high-level task descriptions into lower level control actions.

Training VLA models from scratch is highly resource intensive and is therefore largely restricted to well-funded industrial or institutional efforts. As a result, many existing VLA models are trained on a limited set of robot platforms, which constrains their out-of-the-box applicability to a narrow range of hardware and task settings. To address this limitation, a common paradigm is to fine-tune pre-trained VLA models on smaller task or robot specific datasets. This approach enables downstream users to adapt general-purpose VLA models to their own use cases with modest data collection and computational requirements, and has shown to be effective in practice. Recent open-source VLA models, such as SmolVLA from HuggingFace [3], have further lowered the barrier to entry by providing pre-trained models that are substantially smaller in scale than large industrial VLA systems, making them suitable for fine-tuning with modest computational resources.

3 Hypothesis

We hypothesize that in low-data real-robot regimes with partial observability, naive fine-tuning of pretrained visuomotor policies can *decrease* performance by overfitting to unreliable visual cues or viewpoint-specific artifacts. We further hypothesize that observed failures will be explainable through (i) occlusion-driven ambiguity, (ii) dataset noise and limited visual diversity, and (iii) weak grounding between language prompts and action semantics.

4 Materials and Methods

4.1 Task and Setup

All experiments are conducted on the XLeRobot platform, a low-cost, fully open-source dual-arm mobile robot designed for real-world manipulation tasks. The arms are the LeRobot SO101 arms. For this project, we use a single-arm manipulation setup in which the robot performs a pick-and-place task using its right arm: grasp a cube from a tabletop workspace and place it into a designated target region. The arm provides proprioceptive feedback in the form of joint positions and supports teleoperation for demonstration collection. Visual observations are captured using a forward-facing monocular RGB camera mounted on the robot's head.

4.2 Training Data

Demonstrations are collected following the LeRobot v3 dataset format [2], which aligns video observations with robot state and action logs. The dataset comprises approximately 60 success-only pick-and-place demonstrations collected via teleoperation. Episodes are manually started/stopped to capture intentional demonstrations and reduce irrelevant motion. Despite manual filtering, the dataset exhibits limited visual diversity and frequent occlusions of the cube by the arm/gripper during execution.

4.3 Model

We use SmolVLA, an open-source Vision–Language–Action model released by Hugging Face, as our base policy. SmolVLA is a compact VLA with roughly 450 million parameters, designed to be trained on community-collected robotics data and fine-tuned on modest computational resources such as a single GPU or CPU, lowering the barrier to entry for smaller research groups [3, 7]. The model combines a pretrained vision–language backbone (SmolVLM-2) with a transformer-based action expert trained using a flow-matching objective to predict sequences of continuous control actions [8, 9]. Its reduced scale and efficient architecture make it substantially smaller than many other modern VLAs, enabling faster training and inference while maintaining competitive performance on manipulation benchmarks.

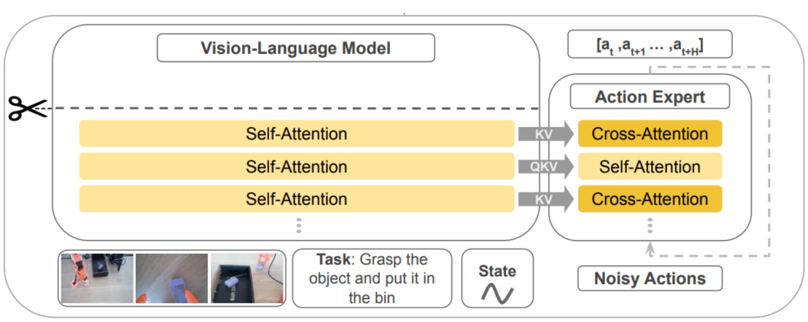

Figure 1: SmolVLA architecture

4.4 Fine-Tuning Procedure

The pre-trained SmolVLA model was fine-tuned on the collected SO-101 dataset. During fine-tuning, the model receives visual observations and a language description of the task as input and predicts continuous control actions for the robot. Training was performed on in Google Colab on a single L4 GPU using modest computational resources, demonstrating the feasibility of adapting VLA models in non-industrial settings.

4.5 Evaluation

We evaluate the fine-tuned policy by executing it on the physical SO-101 robot and qualitatively assessing task completion and behavior consistency. Due to the limited scale of the dataset and the scope of this class project, evaluation focuses on successful task execution rather than large-scale quantitative benchmarks.

5 Results

We evaluate both the base SmolVLA policy and fine-tuned variants on the pick-and-place task.

Base SmolVLA policy. Before fine-tuning, the pretrained SmolVLA policy consistently opened the gripper, reached toward the cube, and positioned the gripper near a plausible pickup location. However, the gripper did not successfully close onto the cube. We attribute this primarily to occlusion: when the arm/gripper entered the workspace, the cube frequently became partially or fully out of view, leaving the policy without sufficient visual evidence to determine grasp success. With monocular RGB-only input, the policy could not reliably infer whether the cube was in the gripper, and often remained stuck in a downward-reaching posture.

Fine-tuned SmolVLA policy. After fine-tuning on the success-only dataset, performance frequently degraded relative to the base model. Across multiple checkpoints, the dominant failure mode was repetitive downward reaching with limited corrective motion or grasp completion. We observed one partially successful trial at the final checkpoint in which the gripper contacted the cube but did not close and halted execution. This behavior required careful action scaling (reduced shoulder pan influence, moderate shoulder lift scaling, constrained elbow/wrist motion, and inverted gripper sign to address an open/close convention mismatch). The behavior was not reliably reproducible; under similar conditions, partial success occurred roughly 30% of the time and was sensitive to visual noise and initial pose.

Action scaling and prompt engineering. Adjusting action gains changed the speed and magnitude of motion but did not alter the qualitative intent: even with large gains, the arm tended to move along a downward trajectory. Sign inversions similarly failed to produce consistent improvements, suggesting the fine-tuned policy was biased toward a particular action pattern. We also experimented with more explicit prompts (e.g., step-by-step instructions describing elbow extension and gripper actions). These prompts did not improve performance and sometimes degraded it, consistent with a lack of grounding between detailed language tokens and the sensorimotor signals available during training.

Overall, these results suggest that fine-tuning on a small, noisy, success-only dataset under partial observability can cause a pretrained policy to overfit to unreliable cues and regress to degenerate behavior.

6 Discussion

Our findings support the view that fine-tuning is not inherently beneficial in low-data real-robot settings. When observations are ambiguous (occlusion-heavy monocular RGB) and demonstrations are viewpoint-misaligned or noisy, supervised fine-tuning can amplify spurious correlations and push the policy toward a brittle, high-probability action mode (e.g., repeated downward motion). This aligns with known failure patterns of behavior cloning under distribution shift [1], but manifests here as a practical obstacle in VLA fine-tuning.

The key takeaway is that the observed failures are actionable. Improving observability (depth, tactile/gripper state, or camera placement), ensuring training and deployment viewpoints match, and adding explicit recovery supervision are likely higher-leverage interventions than additional hyperparameter sweeps. In other words, our constrained setting is a useful diagnostic: it isolates failure causes that can be addressed by concrete data and sensing changes.

7 Limitations as a Controlled Diagnostic Study

This project intentionally uses a constrained setup to make failure modes *interpretable*. Rather than treating the constraints as purely negative, we frame them as a controlled diagnostic study with clear lessons.

Controlled sensing. Using a single fixed monocular RGB camera isolates the role of partial observability and occlusion in real-robot imitation learning. The resulting failures directly motivate sensor or viewpoint upgrades (e.g., depth or gripper-state feedback) as targeted next steps.

Small-data regime. Training on a small success-only dataset reflects realistic constraints in many labs and course settings. This regime makes it easier to observe when fine-tuning harms performance and to attribute regressions to data noise, limited diversity, and viewpoint mismatch.

Actionable failure analysis. Although we do not provide large-scale quantitative benchmarks, the qualitative analysis reveals consistent failure patterns and sensitivity to control scaling and prompt choices. These findings provide concrete guidance for future iterations: add visual diversity, include failure/recovery trajectories, align demonstrations with deployment viewpoint, and incorporate minimal additional sensing to disambiguate grasp success.

Acknowledgments

TODO

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